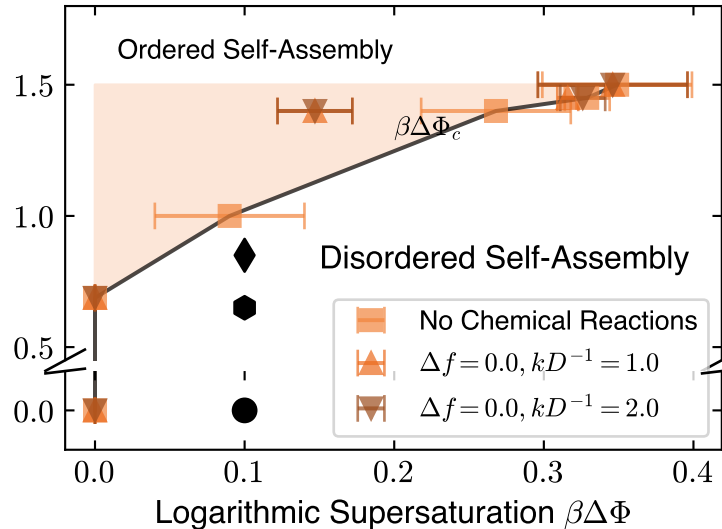


Binding Strength Difference
 $\beta(\epsilon_{jj} - \epsilon_{jl})$

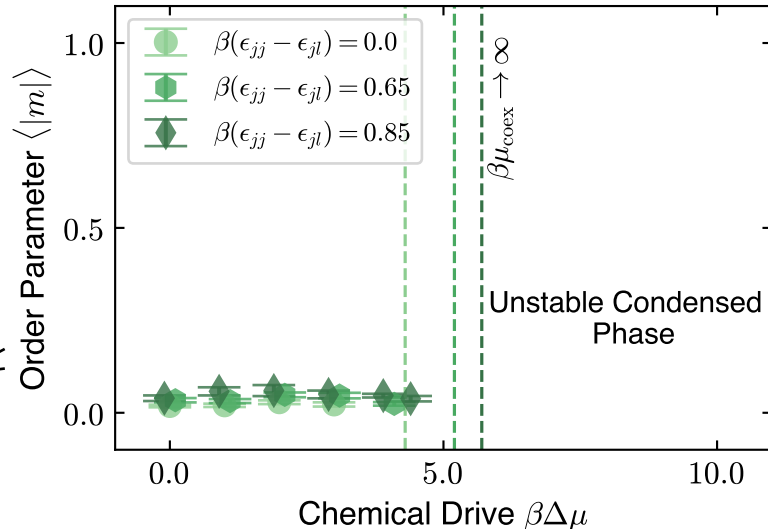
A

Numerical Phase Diagram for Reversible Chemical Reactions



B

Homogeneous Chemical Reactions $\mathcal{S}1$
 $\beta\Delta\Phi \approx 0.1$



C

Heterogeneous Chemical Reactions $\mathcal{S}5$
 $\beta\Delta\Phi \approx 0.1$

